

The notional category of evidentiality

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Abstract This paper is a focused investigation of analytical options proposed for evidentiality in recent formal literature (world-based, commitment-based, event-based) as well as the empirical motivation behind those options. Much work has been devoted to the ontological relationship between evidentiality and modality, and the paper pays special attention to the types of data that motivate the analysis of evidentials as intensional operators. My ultimate goal is to scrutinize the dialectic. While various arguments have been used to adjudicate between the existing theories, I (i) highlight that many of these arguments don't in fact settle the debate, and (ii) bring to the fore new types of diagnostics that might, if applied consistently. My hope is for the reader to emerge with a better understanding of evidentiality as a semantic category—and with many directions for future work, if they are so inclined.

Keywords cross-linguistic semantics, evidence in language, evidentiality, indirectness, modality, semantic theory, semantics/pragmatics division of labor

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. Acknowledgments to be added later.

And as the evidence, derived from witnesses and human testimony, is founded on past experience, so it varies with the experience, and is regarded either as a proof or a probability, according as the conjunction between any particular kind of report and any kind of object has been found to be constant or variable. There are a number of circumstances to be taken into consideration in all judgments of this kind; and the ultimate standard, by which we determine all disputes, that may arise concerning them, is always derived from experience and observation.

—*Of Miracles*, David Hume

1 Introduction

One of the fundamental questions in formal semantics is how to model meaning. Semantic categories are instrumental in addressing this question, as they help us to group linguistic expressions into natural classes and to make predictions about their behavior. Inspired by Angelika Kratzer’s classic papers on modality (1977; 1981; 1991), this chapter seeks to delineate the nature of evidentiality as a semantic, rather than morphosyntactic, category through a focused investigation of analytical options proposed for evidentials in recent formal literature as well as the empirical motivation behind those options.

Much work has been devoted to the ontological relationship between evidentiality and epistemic modality: are they one category? different? distinct but overlapping ones? The answers vary. According to the prominent view spearheaded by Roumyana Pancheva in (Izvorski 1997), evidentials can be treated as modals within Kratzerian semantics (Section 3.1). According to another prominent view put forth by Martina Faller (2002; 2012; 2019) and Sarah Murray (2010; 2014; 2017), evidentials should be treated as operators interacting with the structure of speech acts (Section 3.2). Both of these views are semantic: they treat the evidential signal as hard-wired, and, despite major conceptual differences between them, often make similar empirical predictions. Alongside the semantic doctrine, there exists a somewhat less dominant pragmatic view whereby evidential effects are derived through implicature reasoning (Section 3.3).

My ultimate goal is to scrutinize the dialectic. While various arguments have been used to adjudicate between the existing theories, I (i) highlight that many of these arguments don’t in fact settle the debate and (ii) bring to the fore novel diagnostics that might, if applied consistently. I pay special attention to the types of data that motivate the analysis of evidentials as intensional operators. My hope is for the reader to emerge with a better understanding of evidentiality at the semantics-pragmatics interface—and with many directions for future work, if they are so inclined.

The paper is structured as follows. Section 2 briefly describes the empirical landscape. Section 3 outlines three families of approaches to evidentiality, with a focus on implementation. Section 4 discusses ways of diagnosing intensionality. Section 5 ponders over arguments in favor of hard-wiring discourse-sensitivity. Section 6 proposes a way to test for semantic indirectness. Section 7 concludes.

2 Background

Within typology, evidentials are traditionally understood as expressions that encode the source of information for an utterance (Chafe and Nichols 1986; de Haan 2013a). Based on many languages with evidential paradigms obligatorily expressing how the speaker acquired the prejacent, Aikhenvald (2004) establishes evidentiality as an independent grammatical category that isn't a species of modality (contrary to what was previously assumed; Bybee 1985; Palmer 1986; van der Auwera and Plungian 1998). Aikhenvald's explicit goal is to separate genuine evidentials from other categories that may have an evidential flavor.¹ However, the nature of *semantic* categories is more elusive. For example, Bittner (2014) argues that temporality (semantic category) can receive a unified treatment in languages with or without tense (grammatical category). To this end, it's important to furnish semantic arguments in favor of a particular approach to evidentials, and this is precisely what the bulk of this paper is devoted to.

Following Willett (1988), evidence types relevant to evidentials are often grouped into two broad classes (see also de Haan 2013b): **direct**, often sensory, evidence (e.g., visual or auditory) vs. **indirect** evidence (hearsay and inferential).² We can consider expressions that convey those distinctions to be evidential in some intuitive sense. And just like modal meaning can be expressed through a variety of morphosyntactic means (Arregui et al. 2017), evidential meaning surfaces in many areas of grammar (Diewald and Smirnova 2010b; Forker 2018a; Johanson and Utas 2000; Peterson et al. 2010). Take English. Despite lacking evidentials in the narrow sense, it has a host of expressions that have been argued to exhibit evidence-based restrictions (cf. also Bary and Korotkova 2023:2): *according to* (Kaufmann and Kaufmann 2020; Ziegeler 2023), *apparently* (Krawczyk 2012), *find* (Korotkova and Anand 2021), *find out* (Korotkova and Anand 2024), *must* (von Fintel and Gillies 2010), *looks like* (Rudolph 2023), *tasty* (Anand and Korotkova 2018), *see* + clausal complements (Whitt 2011), *supposed to* (Bochnak and Csipak 2015), *will* (Winans 2016).

Another example of evidentials that do not form a separate grammatical category comes from Georgian (South Caucasian: Azerbaijan, Georgia), a language that optionally marks evidentiality in the past tense paradigm (Boeder 2000; Topadze 2011). It has a neutral past (1a), which is indiscriminate with respect to evidence type, and an evidential past (1b),³ which is only compatible with hearsay and inferences based on visual information.⁴

(1) Georgian

Context 1 (direct): You observes a blizzard.

Context 2 (visual inference): You wake up to white-colored streets.

Context 3 (hearsay): You read a weather report.

1. See Boye (2010); Diewald and Smirnova (2010a) for criticism of this rigid view from a functional-typological perspective.

2. While the direct-indirect distinction has been useful to provide a taxonomy of evidence types, it isn't always fine-grained enough to capture some conceptual underpinnings of evidence in language, such as the types of reasoning evidentials refer to (Davis and Hara 2014; Krawczyk 2012; Takubo 2009; Winans 2016) or part-whole relations between situations they may encode (Hirayama 2020). Furthermore, it is empirically inadequate for cases where a single expression may be compatible with direct or indirect evidence depending on context (Kalsang et al. 2013; Lee 2013; Matthewson 2011; Smirnova 2012).

3. The morphological form in (1b) is synchronically ambiguous between the evidential past and the resultative perfect, the adverbial 'just' is only compatible with the former. There is no designated evidential/perfect morpheme, as Georgian encodes temporal and aspectual information through stem choice (Aronson 1991).

4. Unless indicated otherwise, the data come from my work with native speakers. Glosses will be added later.

- a. Context 1: ✓, Context 2: ✓, Context 3: ✓
 axlanxan tovl-i **movida** [neutral past]
 just snow-NOM come.PST
 ‘It just snowed.’
- b. Context 1: #, Context 2: ✓, Context 3: ✓
 axlaxan tovl-i **mosula** [evidential past]
 just snow-NOM come.IND.PST
 ‘It just snowed, as I hear/infer.’

Across languages, evidentials have much in common and we expect any theory to capture those similarities.

Discourse status Evidential meaning is typically not-at-issue (background information that is not part of the assertion; Potts 2005; Tonhauser et al. 2013), often analyzed akin to presuppositions or conventional implicatures (see discussion in Korotkova 2020).⁵ Capitalizing on their similarities, Murray (2010, 2014, 2017) treats evidentials on a par with slifting parentheticals and apositives in languages like English (though these constructions do not always pattern together; e.g., (16)). Not-at-issue entailments, including presuppositions, often project, however, the projective behavior of evidentials varies across languages and environments. As Korotkova (2020) discusses in detail, the best diagnostic of evidential not-at-issueness is answerhood, the ability to address the QUD via answering explicit or implicit questions (Simons 2007; Tonhauser 2012). As (2), only the prejacent proposition is QUD-relevant. For the purposes of this paper, I assume that evidentials contribute not-at-issue information without delving into the specifics and translate evidential restrictions using parentheticals for simplicity.

(2) Georgian

#Question 1: *What makes you think there is a new metro line in LA?*

✓Question 2: *Any news on public transportation in LA?*

los-angeles-ʃi metro-s axal-i haz-i **gauxvaniat**
 LA-in metro-GEN new-NOM line-NOM construct.3PL.IND.PST
 ‘They built a new metro line in Los Angeles, as I hear/infer.’ (Korotkova 2020:4)

Belief In addition to encoding evidence type, evidentials typically give rise to inferences about the strength of belief said evidence produces (as noted already in Chafe and Nichols 1986): non-hearsay evidentials, just like doxastic attitude predicates (*conclude*, *gather*, *guess*, *think*), require the speaker to consider *p* a live possibility.⁶ Thus, statements of the

5. Evidentials have also been treated as triggers of use-conditional meaning within multidimensional frameworks (McCreedy 2010; van Elswyk 2023), however, I am not aware of any empirical evidence establishing that evidential content is truly use-conditional rather than just parenthetical.

6. According to Aikhenvald (2004:4), “Linguistic evidentiality has nothing to do with [...] indicating one’s belief. All evidentiality does is supply the information source”. While this may be true for hearsay evidentiality (Section 5.1), Aikhenvald’s view predicts the existence of non-hearsay—e.g., visual or inferential—evidentials compatible with a complete lack of belief. This prediction is not borne out: extensive research within formal semantics has not uncovered evidentials that would be used in the counterparts of *I saw it but I don’t believe it* (sensory evidentials are usually banned in cases when the speaker has reasons to mistrust their perception) or *I am in a position to make an inference that p but I don’t believe it*. This shows that there is a deep connection between the encoding of information source and belief.

form ‘Ev ϕ but I don’t believe ϕ ’ are contradictory (Murray 2017:17-21). (3) illustrates. The impossibility to cancel belief with evidentials is typically analyzed semantically (see discussion in van Elswyk 2023), in parallel to similar data on epistemic modals (Yalcin 2007) and in contrast to the standard pragmatic analysis of Moore-paradoxical assertions *Maria is crying but I don’t believe it* (see discussion by Gillies (2001), who also offers a semantic solution).

(3) Georgian

Context: You see Maria’s red eyes and conclude that she was crying. Then you realize that she must have hay fever.

#maria-s utiria magram asi ar aris, es allerg-is-gan
 Maria-DAT cry.3SG.IND.PST but this NEG be.3SG.PRES s/he allergy-GEN-BECAUSE
 akvs tval-eb-i c’itel-i.
 have.3SG.PRES eye-PL-NOM red-NOM
 Intended: ‘Maria was crying, I infer, but this is not so, she has red eyes because of an allergy.’

It is worth asking whether those varied constructions, in Georgian, English and in other languages, may have a common semantic core. In this paper, my key concern is modelling: which semantic mechanisms are useful in analyzing evidential meaning? Formal-semantic literature often understands evidentials as means to provide grounds for claims (see references and discussion in Bary and Korotkova 2023; McCready 2015), and offers several competing views to cash out this intuition. As I will emphasize throughout the paper, my goal is not to advance a particular view but to closely examine the types of arguments that can be used to support or refute a theory. The next section lays out the groundwork for this discussion, but before I proceed, two caveats are in order.

First, with the exception of the interaction between evidentials and the speaker’s commitment, I will not discuss the role of evidentials in conversational information flow or discourse at large, such as their rhetorical functions (Berlin and Prieto-Mendoza 2014; Mushin 2000; Xu 2022) or their effects on the speaker’s perceived credibility and reputation (Krifka and Modarresi 2023; McCready 2015).

Second, I will not discuss what ‘evidence’ amounts to. Some accounts treat updates with evidentials as special cases of probabilistic learning (Davis et al. 2007; McCready and Ogata 2007; McCready 2015 and especially Krawczyk 2012). However, Bayesian epistemology is only one framework to approach evidence and justification (Weisberg 2021) and not an uncontroversial one (Glymour 2016), so I leave it to future research to see which theory is best suited for evidential grounding in language.

3 Theoretical landscape

This section outlines three families of approaches to evidentiality within formal semantics and pragmatics: world-based, commitment-based, and event-based. My goal is to give the reader an intuition about what each type of approach captures, without necessarily staying faithful to the original implementation.

3.1. World-based accounts

Starting with [Izvorski's \(1997\)](#) work on Bulgarian evidential perfect, world-based accounts typically analyze evidentials within the formal apparatus of Kratzerian semantics for modality,⁷ and are often referred to as ‘the modal view’ on evidentiality. Accounts of this general type have been proposed for evidential expressions in many languages, including Cuzco Quechua (Quechua: Peru; [Faller 2011](#)), Gitksan (Tsimshianic: British Columbia, Canada; [Peterson 2010](#)), Japanese (isolate: Japan; [Hirayama and Matthewson 2022](#); [Matsubara 2017](#)), Korean (isolate: Korea; [Lee 2013](#)), Nuu-chah-nulth (Wakashan: British Columbia, Canada; [Waldie 2013](#)), St’át’imcets (Salish: British Columbia, Canada; [Matthewson et al. 2007](#); [Matthewson 2011](#)), Standard Tibetan (Tibeto-Burman: Tibet Autonomous Region; [Garrett 2001](#)), Tagalog (Austronesian: Philippines; [Schwager 2010](#)) and Turkish (Turkic: Turkey; [Meriçli 2016](#); [Şener 2011](#)). Additionally, as the research on evidentiality made its way into semantic mainstream, many expressions traditionally classified as modals have been also argued to carry an evidential signal, including English epistemic *must* ([von Fintel and Gillies 2010](#); [Lassiter 2016](#)) and German *sollen* ([Ehrich 2001](#)).

To give a précis of world-based accounts of evidentiality, I will use [von Fintel and Gillies's \(2010\)](#) framework for *must*. It introduces kernels, special modal bases to represent direct knowledge. In order for a proposition to be known directly and thus be part of the kernel, the kernel must directly settle either this proposition or its complement.⁸ For any given kernel K , $\bigcap K$ represents all knowledge, both direct and indirect, and is a Kratzerian epistemic modal base.

- (4) a. A kernel K is a set of propositions that are known directly.
- b. The proposition $\bigcap K$ is a vanilla epistemic modal base, f_{ep} : the set of worlds compatible with what is known directly and indirectly.
- c. Kernel K directly settles ϕ iff $\exists q \in K [q \subseteq \phi \vee q \subseteq \neg\phi]$.

We can now model direct (5) and indirect (6) evidentiality using kernels, the latter being [von Fintel and Gillies's \(2010\)](#) analysis of English *must*.⁹

- (5) $\llbracket \text{DIRECT } \phi \rrbracket^{c,g,w}$
 - a. At-issue: $\forall w' \in f. \llbracket \phi \rrbracket^{c,g,w'}$
 - b. Not-at-issue: $f = \bigcap K$, K directly settles $\llbracket \phi \rrbracket^{c,g,w}$

7. The dynamic accounts in ([Kierstead 2015](#); [McCready and Ogata 2007](#)) and [Korotkova's \(2016\)](#) analysis of evidentials as attitude reports both belong to this group, as they treat evidentials as intensional.

8. What counts as ‘directly settles’ may vary. Some direct evidentials require sensory grounding (Tibetan; [Garrett 2001](#); [Kalsang et al. 2013](#)), while some have been argued to require the best possible evidence for a given proposition, which includes common knowledge in addition to perception (Cuzco Quechua, [Faller 2002](#); Southern Aymara, [Martínez Vera 2024](#)). The same goes for ‘doesn’t directly settle’. Some indirect evidentials impose specific constraints on the evidence type (e.g., Georgian evidential past is incompatible with inferences based on non-visual information), while some seem to require second-best evidence (Turkish; [Meriçli 2016](#)) or the absence of directness ([von Fintel and Gillies \(2010\)](#) on *must*, [Frana and Menéndez-Benito \(2023\)](#) on Italian future); cf. footnote 2. The framework can handle these subtle differences by fine-tuning the nature of modal bases in question, and my main goal here is to illustrate its general principles.

9. World-based accounts typically treat the evidential restriction as a presupposition, but nothing in the framework prevents us from treating it as a novel not-at-issue contribution, in line with much recent literature.

- (6) $\llbracket \text{INDIRECT } \phi \rrbracket^{c,g,w}$
- a. At-issue: $\forall w' \in f. \llbracket \phi \rrbracket^{c,g,w'}$
 - b. Not-at-issue: $f = \cap K$, K does not directly settle $\llbracket \phi \rrbracket^{c,g,w}$

In words, both direct and indirect evidentials can be treated as universal quantifiers over epistemically accessible worlds. They differ in whether the kernel directly settles the prejacent. For example, if the kernel contains the proposition ‘there is a blizzard’, as in Context 1 in (1), then the prejacent of the sentences in (1), ‘it snowed’, is directly settled, since blizzard entail snow, which makes (1a) acceptable and (1b) infelicitous. While most world-based accounts have been proposed for indirect evidentials, (5) resembles the proposal in (Şener 2011:75) and essentially treats direct evidentials as veridical operators with an additional restriction on the nature of evidence (cf. Korotkova and Anand 2024 on *notice* vs. *know*).

$\cap K$ may entail propositions that aren’t entailed by any q in K : this is indirect knowledge. Such propositions can be jointly entailed by a conjunction of propositions known directly or indirectly ($\cap K$ is closed under entailment). Thus, in Context 2 the kernel can be taken to contain such propositions as ‘the streets are white’ and ‘snow colors streets white’, which jointly entail that it snowed, so the evidential past is felicitous. While a departure from Izvorski’s (1997) original proposal for Bulgarian, this treatment preserves the spirit of her approach insofar as evidentials quantify over worlds that represent knowledge.¹⁰

Let’s take stock. The account sketched above treats evidentials as conventional triggers of not-at-issue content and straightforwardly captures the belief inference. These are welcome results, but there are also potential issues.

According to a common line of attack, certain evidential expressions don’t pattern like *must* and *might* and therefore shouldn’t be analyzed as Kratzerian modals. Izvorski herself doesn’t argue that evidentials are epistemic modals, only that they can be analyzed using the same framework that is often applied to modality. Later research, however, often conflates implementation and categorial status:

- because direct evidentials, but arguably not epistemic modals, express full certainty, they have been argued to be non-modal (Faller 2002), even though nothing precludes the analysis along the lines of (5) and in fact epistemic *must* has been argued to encode full certainty (von Stechow and Gillies 2010, 2021; Mandelkern 2019);
- because hearsay evidentials, unlike *must* and *might*, allow for the lack of belief, they have been argued to be non-modal (Faller 2002, 2019; Murray 2010), even though the data can be analyzed within the Kratzerian framework (Section 5.1);
- because inferential future doesn’t pattern like markers of epistemic necessity or possibility (Winans 2016), it has been argued to be non-modal at least in some languages (Frana and Menéndez-Benito 2023), even though it can be analyzed as non-epistemic modality (Cariani 2021; Cariani and Santorio 2018; Willer 2022).

My larger point is as follows. Possible-world semantics offers much flexibility to account for modality in natural language. Just like modals can be epistemic or circumstantial (flavor), universal or existential (force), evidentials, too, vary in the type of evidence they lexicalize and the strength of belief this evidence produces. The system allows us to capture these distinctions by manipulating quantificational force and conversational backgrounds

10. Some of the world-based accounts of evidentiality inherit the issues related to epistemic modality in that they don’t make a distinction between knowledge and (justified) belief. I put it aside here.

(cf. [Kratzer 2012](#); [Matthewson 2020](#)), which brings us to the fundamental question: how to show whether or not evidentials are modal in some sense?

I will understand modality in a purely formal fashion as involving intensional domains (cf. [Vetter 2011](#)).¹¹ From this vantage point, many expressions belonging to different syntactic categories are modal, including attitude predicates ([Hintikka 1969](#)), degree modifiers ([Heim 2000](#)), epistemic indefinites ([Alonso-Ovalle and Menéndez-Benito 2015](#)), intensional verbs with object DPs ([Forbes 2020](#)) or the progressive ([Landman 1992](#)). To this end, my goal is not to discern whether evidentiality is the same category as epistemic modality (it clearly isn't in some cases), but to evaluate to what extent evidentiality can be understood as involving the same semantic mechanisms as modals. As discussed in Section 4, in order to abandon a modal analysis wholesale one has to demonstrate that evidentials do not make reference to an intensional operator in their semantics.

3.2. Commitment-based accounts

First introduced for Cuzco Quechua evidential enclitics ([Faller 2002](#)) and Cheyenne illocutionary mood (Algonquian: Montana, USA; [Murray 2010](#)), this view on evidential meaning is usually called 'illocutionary', as it focuses on how evidentials interact with the structure of speech acts and the speaker's communicative intentions. However, as discussed in detail by [van Elswyk \(2023\)](#), only the proposal in ([Faller 2002, 2012](#)) (as well as [Faller 2024](#), published after van Elswyk's paper) is truly illocutionary, as it models evidentials as modifiers of sincerity conditions within classic speech act theory. Such strictly illocutionary view is dispreferred both on empirical and conceptual grounds, and given the lack of evidence for this approach, I will not discuss it further. Other approaches often categorized as illocutionary are in fact semantic, as they conventionally encode the discourse effects of evidentials and often use dynamic frameworks that blur the distinction between semantics and pragmatics in the first place. It is those approaches that I discuss in this section.

The goal of commitment-based approaches to evidentiality is to model the changes evidential utterances inflict on the conversational scoreboard, something that the accounts sketched in Section 3.1 above don't aim for explicitly. The key idea is that the speaker isn't merely expressing a belief about the likelihood of the prejacent in light of some body of evidence, but also makes certain discourse commitments, which influences both the type of update their utterance is intended to produce and the range of discourse moves available to the addressee.

Below I will use [Farkas and Bruce's \(2010\)](#) Table model to represent updates with evidentials, inspired by the accounts of conversational dynamics of evidential utterances proposed for Bangla (Indic: Bangladesh, India; [Bhadra 2020](#)), Cuzco Quechua ([Faller 2019, 2023](#)), Japanese ([Northrup 2014](#)), Southern Aymara (Andean: Bolivia, Chile, Peru; [Martínez Vera 2020, 2024](#)), and Turkish ([Meriçli 2015](#)). Nothing hinges on this implementational choice—several other accounts can be categorized as belonging to this family despite their diverging assumptions about the architecture of discourse ([AnderBois 2014; AnderBois 2017; Davis et al. 2007; Koev 2017; Krifka 2019; Frana and Menéndez-Benito 2023; Murray 2010, 2014, 2017](#)).

The Table model is a framework that represents discourse interactions via changes in the conversational scoreboard ([Farkas and Bruce 2010; Malamud and Stephenson 2015; Rudin 2022; Scheffler and Malamud 2023](#)). It minimally consists of the sets TC_{Sp} and TC_{Ad}

11. I will not discuss probabilistic approaches to modality that dispense with worlds altogether ([Lassiter 2017](#)).

for individual discourse commitments of the interlocutors (sets of propositions whose truth the agent is committed to; [Gunlogson 2003](#)), the Table for issues under discussion (the QUD stack; [Ginzburg 2012](#), [Roberts 1998/2012](#)), and the common ground cg modeled as the set of joint commitments.¹² Additionally, the model can track evidentiary grounds for the interlocutors' commitments, which is what evidentials have been argued to tap into ([Northrup 2014](#); [Faller 2019, 2023](#)). We can now model assertions with direct (7) and indirect (8) evidentials using this system; TC_x stands for the set of propositions whose truth x is committed to, AeC_x for the set of propositions for which x has adequate evidence, and $IndC_x$ for the set of propositions for which x has indirect evidence (if need be, other sources of evidence can be included).

(7) Update with direct evidentials: proposal, acceptance

K_1	Speaker	Table	Addressee	$\rightarrow K_2$	Speaker	Table	Addressee
	$TC_{Sp} \cup \{\phi\}$	ϕ	TC_{Ad}		$TC_{Sp} \cup \{\phi\}$		$TC_{Ad} \cup \{\phi\}$
	$AeC_{Sp} \cup \{\phi\}$		AeC_{Ad}		$AeC_{Sp} \cup \{\phi\}$		$AeC_{Ad} \cup \{\phi\}$
	Ind_{Sp}		Ind_{Ad}		Ind_{Sp}		Ind_{Ad}
cg				$cg \cup \{\phi\}$			

(8) Update with indirect evidentials: proposal, acceptance

K_1	Speaker	Table	Addressee	$\rightarrow K_2$	Speaker	Table	Addressee
	$TC_{Sp} \cup \{\Box\phi\}$	$\Box\phi$	TC_{Ad}		$TC_{Sp} \cup \{\Box\phi\}$		$TC_{Ad} \cup \{\Box\phi\}$
	AeC_{Sp}		AeC_{Ad}		AeC_{Sp}		$AeC_{Ad} \cup \{\Box\phi\}$
	$Ind_{Sp} \cup \{\phi\}$		Ind_{Ad}		$Ind_{Sp} \cup \{\phi\}$		Ind_{Ad}
cg				$cg \cup \{\Box\phi\}$			

In words, assertions with evidentials put the prejacent on the Table, proposing to update the common ground with it, and additionally specify the type of evidence the speaker has for their claim via distinguished commitment types. Utterances with direct evidentials hardly differ from non-evidential assertions (see especially discussion in [Martínez Vera 2024](#)), which are also associated with the requirement to have good-enough evidence for the content proposition (a fact often attributed to Gricean Quality or the special assertion norm; see overviews in [Benton 2016](#); [Simion and Kelp 2020](#)). (7) simply states that the speaker has adequate evidence for their claim. Indirect evidentials often end up having the same effect as hedges, constructions that may weaken the speaker's commitment without affecting the prejacent semantically ([Benton and van Elswyk 2020](#); [Koev 2022](#); [McCready 2015](#); [Murray 2014](#)).¹³ The literature varies in the treatment of inferential evidentials: sometimes such evidentials are argued to have a discourse contribution similar to modals without a modal semantics ([Murray 2010, 2014, 2017](#)), while sometimes they are treated as having a modalized prejacent ([Faller 2002](#)). The representation in (8) follows the latter option. The contribution of (1b) within this system amounts to the speaker's proposing to update the common ground with an equivalent of *It must have snowed*, also signalling that they have indirect evidence for this. If the proposal is accepted, the resulting common

12. The role and nature of the common ground has been discussed extensively in recent literature ([Bary 2025](#); [Lederman 2016](#); [Simons 2025](#)). While I cannot do it justice here, it also opens up new opportunities for thinking about speech acts with evidentials.

13. The original Table framework, including the version presented above, does not distinguish between weak vs. strong commitments per se, though one could argue that degrees of commitment are linked to evidence ([Farkas and Roelofsen 2017](#); [Northrup 2014](#)). One can also incorporate graded commitments directly into the system ([Scheffler and Malamud 2023](#)). See [Rudin \(2022\)](#) for detailed discussion.

ground will include the prejacent proposition, however, this is only one possible outcome and the proposal may be challenged or rejected. This contrasts with evidential restrictions, which, as all not-at-issue content, are taken to enter the common ground directly via non-negotiable, automatic updates (see [Scheffler and Malamud 2023](#) for a more nuanced view on (not-)at-issueness in the Table model and [Murray 2014](#) on updates with evidentials).

Let's take stock. Commitment-based accounts capture evidential not-at-issueness and derive belief qua commitment. Additionally, unlike world-based accounts in their current shape, they model discourse updates associated with evidential utterances. On which grounds can we choose between the two?

Some researchers put forth the heterogeneity view such that there is a genuine semantic variation between 'modal' vs. 'illocutionary' evidentials ([Faller 2002, 2007](#); [Garrett 2001](#); [Matthewson et al. 2007](#); [McCready and Ogata 2007](#); [Peterson 2010](#); [Waldie et al. 2009](#)). Let's consider one argument for this view: the differences in embeddability. Evidentials that can occur in complement clauses have been argued to be propositional operators (e.g., in Japanese or St'át'imcets), while those that are banned from clausal complements have been treated as speech act modifiers (e.g., in Cheyenne or Cuzco Quechua). However, as argued in ([Korotkova 2021](#)), this variation is syntactic: the embeddability of evidentials is connected in a principled way to the availability of suitable embedders. Evidentials often occupy a high structural position (not necessarily fixed across languages, pace [Cinque 1999](#)), and not all types of complement clauses have enough structural space to host them. For example, in Turkish evidentials can appear in tensed complements, but not in nominalizations (which lack syntactic layers above TP; [Borsley and Kornfilt 2000](#); [Kornfilt and Whitman 2011](#)). Cuzco Quechua, on the other hand, only has nominalized complements that have a reduced clausal spine compared to matrix clauses ([Cole and Hermon 2011](#)), which, in turn, explains the non-embeddability of Cuzco Quechua evidential enclitics in purely syntactic terms (cf. also [Martínez Vera 2020](#) on Southern Aymara). [Korotkova \(2021\)](#) proposes that similar explanations are applicable to other languages with non-embeddable evidentials, all of which simply lack complements that would have high projections. The bottom line is that the differences in (non-)embeddability do not support the heterogeneity view on evidentiality. More generally, much of the existing variation can be parameterized within either type of account, and there is little evidence for the existence of two types of evidentiality (see detailed discussion in [Korotkova 2016](#); [Matthewson 2020](#); [Murray 2017](#)).

World-based accounts and commitment-based accounts are not irreconcilable conceptually but rather differ in angle: the former deal mostly with the conventional meaning of evidential expressions, while the latter with their conversational dynamics. Much of the literature focuses on the impact evidentials have in modulating the speaker's commitment and the nature of the resulting speech act. However, epistemic modals also influence the discourses containing them, including the speaker's commitment. A well-known problem for Kratzerian semantics, it has been fruitfully addressed in dynamic frameworks that take the truth-conditional meaning of modal utterances as preconditions for their effect on subsequent information states ([Rudin 2025](#); [Veltman 1996](#); [Willer 2013, 2015](#); [Yalcin 2007, 2011](#), among many others; see overview in [Cariani 2020](#) and also [von Stechow and Gillies 2011](#) for some non-dynamic solutions).

I propose the following way out of this conundrum. First, we need to separately establish the presence of an intensional operator to motivate any world-based account of evidentiality. Second, we need to find data that cannot be handled within world-based accounts enriched with assumptions about modal discourses. I turn to each of those issues in Sections 4 and 5, respectively.

3.3. Event-based accounts

Approaches considered so far are semantic: evidentials are treated as encoding the evidential signal. In this section, I turn to a different family that I will call ‘event-based’ in which evidential restrictions are reduced to event mereology. First proposed by Nikolaeva (1999) for Northern Ostyak (Uralic: Siberia, Russia), these accounts capitalize on the fact that many genetically and geographically unrelated languages have morphology that simultaneously contributes both evidential as well as temporal and/or aspectual meaning (see overviews in Bowler 2018; Visser 2015). This type of marking is especially common in the Anatolia-Balkan-Caucasus region (Johanson and Utas 2000), and Georgian constitutes a good example of such system. Event-based accounts have been proposed for the following languages: Bulgarian (Koev 2011),¹⁴ Cuzco Quechua non-witnessed past (not the same morpheme as evidential enclitics; Faller 2004), Korean (Chung 2007), Standard Tibetan (Kalsang et al. 2013) and Tatar (Turkic: Bashkortostan and Tatarstan, Russia; Bowler 2018).¹⁵ Additionally, English *by*-phrases have been argued to give rise to an indirectness inference while having a purely temporal semantics (Altshuler and Michaelis 2020).

The key idea is as follows. Direct and indirect evidence aren’t semantic primitives, but arise from (spatio)temporal configurations between the eventuality described by the prejacent and the eventuality of evidence acquisition, which such evidentials are argued to make reference to. Within this system, direct evidentiality can be understood as a temporal overlap between the prejacent event’s runtime and the event of evidence acquisition (9). Indirectness, on the other hand, results from a non-overlap (10).¹⁶

$$(9) \llbracket \text{DIRECT } \phi \rrbracket^{c, \langle w, t \rangle, f, g} =$$

- a. At-issue: $\exists t' [\phi(t')]$
- b. Not-at-issue: the speaker acquired ϕ in w at t'

$$(10) \llbracket \text{INDIRECT } \phi \rrbracket^{c, \langle w, t \rangle, f, g} =$$

- a. At-issue: $\exists t' [t' \prec t \wedge \phi(t')]$
- b. Not-at-issue: the speaker acquired ϕ in w at t'' such that $t' \prec t'' \preceq t$

In words, direct evidentiality is a by-product of the fact that the speaker learned the prejacent as the eventuality described by it was unfolding. Different flavors of indirectness, on the other hand, occur when the speaker learned the prejacent after the fact, which excludes having sensory evidence for it but is compatible with inference or hearsay. For example, the (in)felicity of the Georgian evidential past in different scenarios in (1) could be explained as follows. In Context 1, the snowfall and the speaker’s observing it must overlap in time, which would make the evidential past infelicitous (and the semantics of the regular, evidence-neutral, past can be formulated in the standard way, without making reference to evidence acquisition at all). In Contexts 2 and 3, on the other hand, the speaker may acquire evidence after the snowfall took place, which satisfies the evidence acquisition condition without hard-wiring indirectness.

14. The later version in (Koev 2017) presents a hybrid approach wherein not only temporal relations, but also the speaker’s commitment are hard-wired.

15. Davis and Hara’s (2014) causal account of Japanese also belongs to this family as it decomposes evidence into a relationship between events and does not have a doxastic component.

16. This is a much simplified version that ignores many complex patterns that have been argued to play a role in the semantics of event-based evidentials. See references and discussion in Johnson (2022), who analyzes the elaborate past tense system in Mvskoke (Eastern Muskogean: Oklahoma, USA).

Event-based accounts present us with the possible existence of a genuinely different class of expressions. One major difference from both world-based and commitment-based accounts is that in this case, any evidential effects must be calculable off the base semantics and that they can sometimes fail to arise (as is the case for canonical conversational implicatures). For an illustration of how this might work, consider English *by* temporal adverbials with present-tensed (11a) vs. past-tensed predicates (11b).

- (11) a. Shira is home **by now**.
 $\rightsquigarrow$ The speaker isn't certain.
- b. Shira was home **by now**.
 $\nrightarrow$ The speaker isn't certain.

(adapted from Altshuler and Michaelis 2020:526)

Altshuler and Michaelis argue that *by* temporal adverbials require that the onset of the resulting state described by the base predicate must precede the time described by the adverbial, which in turn gives rise to the implicature that the speaker likely didn't witness Shira's entering. Moreover, with present-tensed, but not past-tensed predicates, the speaker isn't fully certain (since otherwise a different expression with a more precise temporal reference would be used). Thus, *by* temporal adverbials do not directly encode evidential restrictions, nor do they convey information about the speaker's doxastic state. We expect putative event-based evidentials to behave in a similar way. For example, the entry in (9) predicts the lack of obligatory belief inference for event-based evidentials, including direct ones—one may observe something happening, but still doubt what they see or hear (as with small clause complements of perception verbs in English; Barwise 1981). The evidential restriction can be cancelled with the perfect in Sanzhi Dargwa (Nakh-Daghestanian: Daghestan, Russia; Forker 2018b), however, this prediction isn't always addressed in the corresponding literature.

One of the hallmarks of event-based accounts is that they are extensional. However, there are other analytical options. There is a deep connection between temporality and modality. On the one hand, canonical modals encode distinct temporal relations (Condo-ravdi 2002), and there is an established tradition of treating future as a modal (Cariani 2021; Enç 1996; Klecha 2014; Werner 2006; Winans 2016). On the other, temporal markers are often co-opted to express modal meanings, such as counterfactuality (Iatridou 2000; Mackay 2019; Schulz 2014). This opens many possibilities for how evidential meaning can be combined with temporal ingredients. Thus, some evidentials have been argued to be essentially relative tenses with a doxastic component (Bulgarian, Smirnova 2012; Korean, Lee 2013; Mvskoke, Johnson 2022). Hirayama and Matthewson (2022) make an even stronger claim and propose to parametrize variation in the temporal constraints of evidential expressions by using the relative ordering between the evidence acquisition time and the onset of the resulting state described by the prejacent. Their main point is that, because specific expressions lexicalize different orderings (just like modals differ in their temporal perspective), evidential restrictions cannot be always calculated pragmatically. In what follows, I will not be discussing reasons for postulating a temporal component in the semantics of evidentials and instead focus on other semantic ingredients.

4 Intensionality

Commitment-based accounts treat evidentials as interacting with general grounding mechanisms for speech acts. Event-based accounts treat evidentials as extensional operators. What they both have in common is that evidentials are analyzed without necessarily making reference to intensional domains, unlike in world-based accounts. As discussed in previous sections, construing evidentials as modal-like elements is not incompatible with assigning them certain pragmatic functions or incorporating temporality in their semantics. To this end, it is important to diagnose whether evidentials induce effects we normally expect from intensional operators. In this section I go over two such diagnostics familiar from the literature on modals and attitudes: modal subordination and mixed ‘de re’/‘de dicto’ readings. The upshot is as follows: commitment-based accounts can in principle handle evidential intensionality, while event-based accounts cannot.

4.1. Modal subordination

Modal subordination is a phenomenon whereby anaphora and other forms of presupposition satisfaction are often blocked in intensional environments unless in the scope of another intensional operator (Roberts 1996, 2021), and is triggered by a variety of modal expressions, including modal auxiliaries, conditionals or disjunction. The fact that *will* and other predictive expressions trigger modal subordination is one of the key arguments for treating them as modal operators (Cariani 2021; Klecha 2014, though see Boylan 2023), showing the insufficiency of both purely temporal approaches rooted in tense logic (Kissine 2008) and purely discourse-based approaches (Frana and Menéndez-Benito 2023). Here I want to emphasize the importance of systematically testing for modal subordination with evidentials, which so far has been done only occasionally (Faller 2014; McCready and Ogata 2007; Lee 2013; Smirnova 2012). Consider Georgian (12).

(12) Georgian

Context: You see gardening tutorials on Nana’s desk, dirty gloves, soil in the basement.

- a. nana-s xe_i daurgia
 nana-ERG tree.NOM plant.IND.PST
 ‘Nana planted a tree, I infer’.
- b. mas-ze_i #dasaxldnen / ✓dasaxlebulan cixv-eb-i.
 in-it inhabit.3PL.PST / inhabit.3PL.IND.PST squirrel-PL-NOM
 ‘Squirrels # inhabited / ✓apparently inhabited it’.

The pronoun can refer to an antecedent in the scope of an evidential only if it itself is introduced within the scope of an evidential, otherwise anaphora is blocked. Compare the data in (12) to the behavior of English *by* temporal adverbials. Although associated with an evidential flavor in the present tense, such adverbials are extensional operators (Altshuler and Michaelis 2020), which correctly predicts the lack of modal subordination (13).

(13) By Friday, Magda submitted a paper_i. It_i was on modality.

All world-based accounts predict that evidentials would exhibit the pattern in (12), as they are treated as intensional operators. Thus, they straightforwardly capture the data for Georgian, however, as far as I know, this prediction has not been explicitly checked for all evidentials for which a world-based account has been proposed.

Event-based accounts predict that evidentials would pattern like *by* temporal adverbials

(13)—see, e.g., Koev (2011, 2017) on Bulgarian,¹⁷ Davis and Hara (2014) on Japanese and Chung (2007) on Korean. However, as discussed by Smirnova (2012), McCready and Ogata (2007) and Lee (2013), evidentials in respective languages give rise to modal subordination and therefore cannot be accounted in purely extensional terms.

Finally, the specific predictions of commitment-based accounts depend on general assumptions about anaphora resolution and presupposition accommodation, which aren't always spelled out. Table-based accounts, such as the one sketched in Section 3.2, predict that evidentials would trigger modal subordination only when the proposition on the Table contains an intensional operator: for example, (8) captures the data in (12) since the issue under discussion is $\Box\phi$. However, some approaches (e.g., Faller 2023; Murray 2014) predict that indirect evidentials should not block anaphora, as they only modulate the speaker's commitment.¹⁸

The bottom line is as follows. Modal subordination is a useful tool in adjudicating between extensional and intensional accounts of evidentials. All event-based accounts are extensional, and as such predict the lack of modal subordination, which is clearly incorrect for several evidentials expressions for which such accounts have been proposed. All world-based accounts are intensional and predict that indirect evidentials may give rise to modal subordination (cf. footnote 18). Commitment-based accounts are somewhat of a grey area, however, they can easily made intensional without sacrificing their core assumptions (unlike event-based accounts). Note that it is a one-way diagnostic: the absence of modal subordination does not preclude an intensional account, while its presence strongly speaks against an extensional account.

17. Based on the data in (i), Koev (2017) incorrectly argues that Bulgarian indirect evidential does not give rise to modal subordination. Let us take a closer look.

(i) Bulgarian

Context: You arrive home and find a cake on the table. You infer that your roommate Ivan made it. Later you say:

Ivan napravi-l torta_i. pro_i be-še ostaven-a na masa-ta v kuxnja-ta.

Ivan make-IND cake pro be-PST left-F on table-DEF in kitchen-DEF

'Ivan made a cake, I infer. It was left on the kitchen table.'

(adapted from Koev 2017:25)

The null pronoun in (i) (Bulgarian is a subject-drop language) refers to an antecedent superficially embedded under the indirect evidential. However, anaphora is felicitous since the indefinite DP *torta* 'cake' scope outside of the evidential: the context states a cake exists in the world of evaluation. A proper instance of modal subordination is (ii), where anaphora is blocked without the evidential:

(ii) Bulgarian

Context: You sees cookbooks on Ivan's desk and a dirty baking sheet in the sink.

Ivan napravi-l torta_i. Tja_i #e / ✓bi-l-a v kuxnja-ta.

Ivan make-IND cake she be.3SG.PRES / be-IND-F in kitchen-DEF

'Ivan made a cake, I infer. It #is / ✓apparently is in the kitchen.'

18. Faller (2019) argues that, because Cuzco Quechua hearsay marker =si does not give rise to modal subordination while the German reportative *sollen* does (Faller 2014), only the latter is a true modal. I propose a different take on the data. *Sollen* tends to convey doubt or distancing. But with =si , as Faller herself notes, the speaker commits to the prejacent (in the absence of explicit disavowals). It is therefore natural that anaphora is allowed. Such cases with =si likely instantiate global accommodation, similar to one of the readings in (i), where the speaker accepts the claim:

(i) a. The author claims that Vulcan_i exists after all.

b. It_i has circled around Mercury for ages without us ever noticing it.

(Roberts 1996:719)

In general, operators differ in how easily they allow local accommodation (Heim 1992) and in whether they obligatorily trigger modal subordination (Klecha 2011). It wouldn't be surprising to find some variation within evidentials as well.

4.2. ‘De re’, ‘de dicto’ and contradictions

Intensional environments, including speech and attitude reports, are known to ameliorate contradictions because they allow different parts of the prejacent to be evaluated with respect to different worlds: expressions in the main predicate position must be read ‘de dicto’ while expressions outside of main predicate position can be read ‘de re’ or ‘de dicto’ (Anand and Korotkova 2022; Farkas 1997; Keshet 2010; Musan 1997a; Schwarz 2012). We can apply this diagnostic to hearsay evidentials as they are the only ones that introduce another perspective (see discussion in Section 5.1). Consider (14).

(14) *Context: You read an article about a very dangerous disease, claimed to be caused by a fungus, while in fact you know that it is a virus.*

a. Bulgarian

Naj-smurtonosnijat virus ne **bel** virus
 SUP-deadly virus NEG be.IND.3SG virus
 ≈ ‘**It’s said** that the world’s deadliest virus is not a virus’.

b. German

Der tödlichste Virus **soll** kein Virus sein.
 DEF.M.SG deadly.SUP virus.NOM.SG **REP** NEG.INDEF virus.NOM.SG be.INF

c. Turkish

Dünya-nın en ölümcül virüsü virüs değil-**miş**.
 world.GEN SUP deadly virus.POSS virus NEG.COP-**IND**

As the data above show, Bulgarian indirect evidential, Turkish *miş* and German *sollen* pattern like standard indirect discourse (e.g., their English translation): they allow apparent contradictions in their scope (14). The same sentences are infelicitous without evidentials, just like the English *The deadliest virus is not a virus*.¹⁹

To the best of my knowledge, such data have never been tested before for evidentials, yet they serve as a useful litmus test for different theories. All world-based accounts handle them in a straightforward way, schematized in (15): the subject DP remains in the scope of the evidential and there is some mechanism that allows it to be interpreted with respect to the higher world (Aloni 2001; Charlow and Sharvit 2014; Ninan 2012; Percus and Sauerland 2003).²⁰

(15) [w_0 I hear [w_1 [w_0 the world’s deadliest virus]_{de re} [is not a virus]_{de dicto}].

Just like with modal subordination, these data are particularly problematic for event-based accounts, which are extensional and therefore predict no ‘de dicto’ readings. Note that this is another argument against the extensional account of Bulgarian in Koev (2017), which incorrectly predicts (14a) to be infelicitous, as both DPs are expected to be evaluated at the same world.

Commitment-based accounts in their current shape predict that there is someone, either the speaker or a third party, who is committed to the prejacent proposition. Without a dedicated mechanism for ‘de re’ construal that is enabled by intensional quantification, we end

19. The English sentence may be non-contradictory with a special quotative intonation (cf. Stec et al. 2016) or with scare quotes around ‘not a virus’. In each case, it will be an instance of mixed quotation (Maier 2014), an intensional environment; cf. discussion in (Anand and Korotkova 2022:623, fn.4).

20. While the exact mechanism is unimportant here, ‘de re’ isn’t reducible to scope (Quine 1956 and much later work), so we can’t treat (14) as the relevant DP scoping outside of the evidential.

up with an undesirable result: a commitment to a contradiction in cases like (14). Thus, accounts that treat evidentials and parentheticals on a par (Murray 2010, 2014, 2017) predict that evidentials would not tolerate contradictions, behaving like slifting parentheticals (16).

(16) ??The deadliest virus is not a virus, I heard.

The bottom line is as follows. Sentences that otherwise would be contradictory are felicitous in intensional environments because they allow different parts of the sentence to be evaluated at different worlds, and are an instance of a more general case of hybrid ‘de re’/‘de dicto’ readings. I propose to use this property to diagnose whether worlds need to be a part of the semantic denotation of hearsay evidentials. World-based accounts predict such readings to be available, and thus correctly capture the data in (14). Event-based accounts predict such readings to not be available because they treat evidentials as extensional operators. Commitment-based accounts are not currently designed to account for such readings, however, they can be modified accordingly (e.g., using the system in Kierstead 2015). While there may be a genuine semantic split between hearsay evidentials that allow hybrid ‘de re’/‘de dicto’ readings and those that don’t (much like there is variation in the level of the speaker’s commitment to the prejacent, cf. footnote 21), it would be good to test those predictions empirically.

This section discusses two empirical diagnostics that help detect intensional operators. Importantly, neither diagnostic applies to direct evidentials. They aren’t expected to trigger modal subordination, because the speaker is certain about the truth of the prejacent, and we can’t test for contradictions, as they only talk about the speaker’s perspective. It may be that direct, but not necessarily indirect, evidentials are extensional (this is Garrett’s (2001) analysis of Tibetan; see also Barwise 1981; Higginbotham 1983 on perceptual reports in English). It also may be that direct evidentials introduce vacuous intensional quantification, as represented in (5) (see Bhadra (2022); Matthewson (2020) for a general discussion of direct evidentials as modal expressions). For those evidentials to which the diagnostics apply, we find evidentials that trigger modal subordination and ameliorate contradictions, like any well-behaved intensional environment would. Such data can be easily handled by any world-based account and cannot be handled by purely event-based accounts, as making the latter intensional runs afoul of the spirit of those approaches. Commitment-based accounts present an interesting case. As stressed throughout the paper, they can be seen as complementary—rather than opposite—to world-based accounts. To this end, while they might not be able to handle the data in their current shape, as in most accounts of this sort evidentials are not treated as intensional operators (to be more precise, non-inferential evidentials aren’t, and the status of inferential ones isn’t clear), there is nothing that prevents them from incorporating intensionality into the semantics of evidentials without sacrificing their core assumptions about the discourse updates associated with evidential utterances. My key point is that it is important to test for the intensional component independently, and I hope to see the diagnostics discussed here applied more systematically in future work.

5 Discourse

As discussed already at the end of Section 3.1, the lack of parallelism between some evidentials and some modals is not an argument for abandoning world-based accounts. In this

section, I go over one case that has generated a lot of attention, disavowals with hearsay evidentials, and show that it doesn't in fact adjudicate between different theories of evidential meaning. More generally, there is little empirical evidence that the discourse effects of evidentials have to be derived in the same way as those of hedges rather than those of (epistemic) modals—this is reflected in [Krifka's \(2019\)](#) framework that collapses the distinction between the two. I conclude this section by discussing the types of data that might motivate hard-wiring sensitivity to discourse structure for at least some evidentials.

5.1. Hearsay

One oft-cited argument in favor of commitment-based approaches is the behavior of hearsay evidentials, which often allow the speaker to explicitly disavow the prejacent. This property is very common and doesn't correlate with the morphosyntactic make-up of evidentials or their semantics.²¹ (17) illustrates the pattern.

- (17) a. Dutch *schijnen* (Germanic: Netherlands; [Koring 2013](#))

Het **schijnt** te sneeuwen
 this seem.REP.3SG.PRES to snow.INF
 maar ik geloof daar niets van
 but I believe.1SG.PRES there nothing of
 'It's said to be snowing but I don't believe it'.

- b. Georgian

Fox News reports that California legalized marijuana.
 kalifornia-s k'anonier-i **gauxdia** marihuan-is gamoq'eneba
 California-DAT legal-NOM make.IND.PST marijuana-GEN usage.NOM
 magram asi ar aris
 but this NEG be.3SG.PRES
 ≈'I hear that California legalized marijuana, but that is not true.'

Utterances like (17) are often taken to be non-assertive speech acts ([Faller 2002, 2019; Murray 2010, 2014](#)), given that unqualified assertions require belief. And the possibility of disavowals has been linked to the presence of a third party whose commitments are being reported ([AnderBois 2014; Faller 2019; Koev 2017; Pancheva and Rudin 2019; Smirnova 2012](#)), which is only possible with hearsay evidentials. There is a number of competing views on the market; my goal here is to vindicate world-based accounts.

Because epistemic modals do not allow flat-out disavowals (18), the literature on evidentiality has argued against attributing evidentials the same semantics as *must*.

- (18) #It must be raining on Ferenginar, but I don't believe it.

However, as pointed out by [Yalcin \(2007\)](#), Kratzer's original semantics incorrectly predicts (18) to be felicitous. To illustrate Yalcin's point for evidentials, (19) recasts the seman-

21. Pace [AnderBois 2014](#), there is lexical variation. In at least three languages of the Pacific Northwest hearsay evidentials require the speaker to consider the prejacent a possibility: Gitksan ([Peterson 2010](#)), Nuuchahnulth ([Waldie 2013](#)) and St'át'imcets ([Matthewson et al. 2007](#)). This is expected, as speech reports in general, and communicative predicates in particular, vary in whether they encode information about the speaker's attitude to the reported content: for example, *say* is neutral, *be right* conveys endorsement ([Anand and Hacquard 2014](#)), while *claim* gives rise to non-endorsement pragmatically ([Korotkova and Anand 2025](#)). Depending on the assumptions, this variation can be parameterized both within world-based and commitment-based accounts (with the exception of ([AnderBois 2014](#)), who has to make ad-hoc stipulations about the availability of certain pragmatic mechanisms to account for hearsay evidentials banning disavowals).

tics for indirect evidentiality given in (6) with minimal modifications that allow disavowals.

- (19) $\llbracket \text{INDIRECT } \phi \rrbracket^{c,g,w}$
- a. Not-at-issue: $f = \cap K$, K does not directly settle $\llbracket \phi \rrbracket^{c,g,w}$
 - b. At-issue: $\forall w' \in \max_{g_{st}(w)}(\cap f(w)). \llbracket \phi \rrbracket^{c,g,w'}$,
 where $g_{st}(w) = \{q \mid q \text{ is normal in } w\}$ (a stereotypical ordering source),
 where $\forall u, v : v <_{g(w)} u$ iff $\{q \mid q \in g(w) \wedge u \in q\} \subset \{q \mid q \in g(w) \wedge v \in q\}$
 and where $\max_{g_{st}(w)}(\cap f_{ep}(w)) = \{w' \in \cap f_{ep}(w) \mid \neg \exists v \in \cap f_{ep}(w). v <_{g_{st}(w)} w'\}$.

(19) is close to [Izvorski's \(1997\)](#) original proposal for the Bulgarian indirect evidential (and in fact she explicitly talks about the differences in commitment between hearsay and non-hearsay). The crucial difference between (6) and (19) is the addition of a stereotypical ordering source g_{st} , a function that maps every world w to a set of propositions that represent the normal course of events in w . Epistemically accessible worlds $\cap f_{ep}(w)$ are ordered so that worlds verifying more propositions from g_{st} are ranked higher. The evidential operator quantifies over the maximal subset of $\cap f_{ep}(w)$, as determined by the ordering source. The ranking may be such that the world of evaluation w is not one of the most normal worlds (for example, if the source of the report is not reliable), which would allow the speaker to disavow the prejacent. This option has been forgotten in the literature on evidentiality.

Another option is to treat hearsay evidentials as having an informational, rather than an epistemic modal base, amounting to the quantification over worlds compatible with the contents of reports $f_{inf} = \{z \mid z \text{ is compatible with what was said in } w\}$ ([Ehrich 2001](#); [Kratzer 2012](#); note also that this is Hintikka semantics for *say*). Unlike knowledge, information isn't veridical, and the world of evaluation needn't be included in the modal base (this is similar in spirit to [Smirnova's \(2012\)](#) analysis of disavowals as *de dicto* reports).

The most recent discussion by [Faller \(2019\)](#) simply assumes that world-based accounts do not predict the range of conversational moves available to utterances with hearsay evidentials. In particular, she argues that the prejacent may be intended to resolve the QUD and update the common ground in those cases when the speaker doesn't void their commitment right away. However, the speaker's communicative intentions don't enter the semantic composition in either of the options presented above and can be calculated based on their attitude towards reliability of the source, something that is undetermined by the semantics (cf. [Davis et al. 2007](#); [McGready 2015](#)). Moreover, the modal semantics predicts that both the prejacent and the quantificational claim can be at-issue, which is the case for *might* and *must* ([von Stechow and Gillies 2010](#); [Roberts 2023](#)), so we expect the prejacent of informational modals to be QUD-relevant and proposal-worthy as well.

Another argument put forth by Faller against a world-based account at least for Cuzco Quechua *=si* is that its evidential restriction is never at-issue, while she assumes, following [Faller \(2007\)](#), that world-based accounts predict otherwise. This is a tricky point. In Kratzerian semantics, conversational backgrounds that determine the interpretation of a modal expression are treated as presuppositional (see recent discussion in [Roberts 2023](#)), an intuition preserved in (5), (6), and (19). Thus, within the classic framework it's never under discussion whether *must* gets an epistemic or a deontic interpretation. [Rudin \(2025\)](#) shows why this might be too strong for epistemic modals, however, if hearsay evidentials are treated as non-epistemic modals (as [Kratzer \(2012\)](#) herself explicitly proposes), the default prediction is that the modal base is never at-issue, which seems to be consistent with Cuzco Quechua data. Moreover, [Faller's \(2007\)](#) own examples on German reportative *sollen*

only show that the source of the report—but not its fact—can be at-issue, so we don’t find major differences between putatively modal vs. non-modal hearsay evidentials after all.

To sum up, the available data do not provide evidence for discarding world-based views on hearsay evidentials. More generally, the literature does not make it clear how the discourse contribution of hearsay markers is different from parenthetical speech reports (Hunter 2016). Given that the pattern in (17) is not limited to evidentials, we would like to have a more general account of disavowals with hearsay across a variety of constructions.

5.2. Conversational function

Commitment-based accounts suggest that evidentials make reference to the architecture of discourse. In order to make an argument of this sort, we would need to show that at least some evidentials have non-trivial discourse functions that cannot be derived from their basic semantics (or to provide strong evidence against treating them as intensional operators; Section 4). The existing work on this is scarce, but promising. For example, Martínez Vera (2024) shows that Southern Aymara *=wa* marks evidence source and information focus, and is therefore infelicitous discourse-initially. And Grzech (2020) demonstrates that Upper Napo Kichwa (Quechua: Ecuador) epistemic enclitics have various common-ground management functions, such as signalling shared vs. private knowledge, and are generally closer to discourse particles in, e.g., German (see overviews in Grosz 2020; Zimmerman 2011) than to canonical evidentials (see Bergqvist (2023) for similar remarks on direct evidentials more generally). (20) illustrates for the enclitic *=tá*, which Grzech argues to be an instance of VERUM: emphasis on the speaker’s certainty in the prejacent especially when challenges are anticipated (see overview in Jordanoska et al. 2023).²²

(20) Napwa Kichwa

- | | | | |
|----|--|---------------------|------------------|
| A. | apa-chi-k-llara | pamba-na | |
| | bring-CAUS-NMLZ=ID.REF | bury-INF | |
| | ‘[So] the midwife herself has to bury [the placenta]?’ | | |
| B. | apa-chi-k-llara | pamba-na= <i>tá</i> | |
| | bring-CAUS-NMLZ=ID.REF | bury-INF= <i>tÁ</i> | |
| | ‘Yes, the midwife DOES have to bury [it] herself.’ | | (Grzech 2020:91) |

Put very roughly, if we could show that evidentials have conversational functions similar to those of discourse particles (e.g., an explicit reference to conversational moves or sensitivity to the QUD structure), this would be a compelling argument for treating evidentials as interacting with the structure of speech acts (see also Northrup (2014) on evidential restrictions of canonical discourse particles in Japanese). However, most research so far has shown that evidentials introduce a kind of not-at-issue, parenthetical meaning that influences the speaker’s commitment, and these properties can be analyzed in a variety of ways. This is similar to the debates on the proper analysis of expressions creating bias in polar questions, which can be treated as directly referring to illocutionary notions or as arising through a combination of their meaning and the pragmatics of questions (see overview in Goodhue forth.). At present, more empirical work is needed.

22. Grzech also notes that Faller (2002) describes a similar function for *=mi*, a marker analyzed as a direct evidential signalling that the speaker has best possible grounds for a proposition. It would be interesting to check if *=mi* might in fact be an instance of VERUM rather than an evidential—or maybe both, see Frana and Rawlins (2025) on some parallels between VERUM marking and evidentials.

6 (In)directness

Some evidential expressions have very specific restrictions, such as sensitivity to a particular channel of information or to the direction of reasoning. However, in some other cases we seem to be dealing with an admittedly vague notion of (in)directness (cf. footnotes 2, 8). Event-based approaches exploit this vagueness and often treat indirectness as arising through implicature reasoning: if one wasn't in the same time and place as the eventuality described by the prejacent, one can only have indirect evidence that it took place. Capitalizing on the research on the fine-grained semantics of attitude predicates, I propose a diagnostic for semantic indirectness: embedding under attitudes that have their own evidential restrictions and ban indirect expressions in their complements.

First, consider English *notice*. As [Korotkova and Anand \(2024\)](#) show, it requires the subject to have sensory evidence for the complement and as such excludes situations that necessitate a different way of evidence acquisition, e.g., inference to the best explanation that is not based on observable results (21).

- (21) *Context: Mordecai always invites Pascal to his favorite Vietnamese café when he submits a paper. Pascal got an invitation earlier today.*
 #Pascal noticed that Mordecai (apparently) submitted/must have submitted a paper.

Second, consider *find*-verbs. Across languages they take complements that talk about matters of opinion, not fact ([Coppock 2018](#); [Kennedy and Willer 2022](#); [Sæbø 2009](#); [Stephenson 2007](#)). They only allow subjective predicates in their complements, many of which impose a firsthand requirement (e.g., *tasty*; [Anand and Korotkova 2018](#); [Ninan 2014](#); [Soria-Ruiz 2024](#)), but even when combined with predicates without those requirements, such as *authentic*, they indicate that the subject formed an opinion based on direct, often sensory evidence (first observed, but not argued for, in [Stephenson 2007](#); see detailed discussion in [Korotkova and Anand 2021](#)).²³ In English such verbs typically take only small clauses, but in those languages there they allow full clauses, such as German (22) and Bulgarian (23), they ban epistemic modals and indirect evidentials in their scope even when those are combined with a subjective predicate. The complements are felicitous stand-alone or under *think*-verbs, and *must* and *might*-modals are allowed with a non-epistemic interpretation (e.g., deontic or circumstantial).

- (22) German
 Magda findet, dass das Essen authentisch ist / #sein
 Magda find.3SG.PRES COMP DEF.N food authentic be.3SG.PRES be.INF
muss.
must.3SG.PRES
 'Magda is of the opinion that the food is authentic / #must be authentic.'
- (23) Bulgarian
 Namiram, če torta-ta e vkusn-a / #bi-**l**-a vkusn-a.
 find.1SG COMP cake-DEF.F be.3SG.PRES tasty-F be-**IND**-F tasty-F
 'I find the cake tasty / #apparently tasty.'

23. *Find* takes abstract complements, which has been used as an argument against the directness account ([van Wijnbergen-Huitink 2016](#)). However, directness isn't always the same as sensory evidence. If we take all predicates to come with grounding requirements ([Cariani 2022](#)), we can formulate the semantics for *find* as requiring best possible grounding (à la [Faller 2002](#)), which would amount to firsthand experience with predicates that allow it.

Using a more sophisticated version of the kernel framework (Section 3.1), Korotkova and Anand propose that the infelicities in (22) and (23) are rooted in an evidential clash. *Find* requires its complement to be directly settled while indirect markers, including epistemic *must* and *might*, require that their prejacent is not directly settled. Jointly, these requirements lead to a semantic contradiction (see Korotkova and Anand 2021:524-526 for full derivation). Importantly, this analysis is indifferent to the strength of *must* and only requires it to carry an evidential signal. With more nuanced restrictions on ‘direct settlement’ (cf. footnote 8), it can easily be extended to *notice* (21).

Building on Korotkova and Anand (2021), I propose to use embedding under *notice* and *find* as a way to disentangle between semantic vs. pragmatic views on evidentiality. Provided that the languages in question have such verbs in the first place and that evidentials are embeddable under doxastic predicates in those languages, we expect that semantically indirect expressions would conform to the pattern outlined above: one can’t *notice* or *find* a proposition for which one has indirect evidence. This is in line with semantic accounts of the indirectness of *must*-modals (von Fintel and Gillies 2010, 2021; Lassiter 2016), Bulgarian indirect evidential (Izvorski 1997; Smirnova 2012) and a range of other expressions discussed by Korotkova and Anand.

The prediction of event-based approaches is, on the other hand, that there shouldn’t be any clash. This prediction is borne out for *by* temporal adverbials under *notice*.²⁴ (24a) is felicitous within a narrative, and (24b) is a vanilla simultaneous reading of the embedded past tense.

- (24) a. Pascal notices that Mordecai submits the revised paper by the deadline. As the journal’s editor, he receives a notification right away.
 b. Pascal noticed that Mordecai submitted the paper by the deadline. As the journal’s editor, he received a notification right away.

In (24), the indirectness implicature of *by now* disappears, as temporal implicatures (e.g., lifetime effects; Musan 1997b) often do in embedded environments and both sentences are felicitous. We expect the same pattern for other evidential expressions whose indirectness has been argued to arise pragmatically. However, this is not what we find. The data in (21) present a problem for Mandelkern’s (2019) view on *must* wherein it does not encode indirectness semantically (see von Fintel and Gillies (2021) for other issues). Universal epistemic modals, including *must*, are in general licensed under doxastic attitudes like *know* or *think* (Anand and Hacquard 2013), so it’s unclear why (21) would be bad. An evidential analysis of *notice*, coupled with an evidential analysis of *must*, explains the data in a principled way. Likewise, if Bulgarian indirect evidential does not actually encode indirectness (as per Koev 2011, 2017), it is unclear why it is infelicitous in (23).²⁵ One could try to furnish an explanation that *find* requires some kind of spatio-temporal overlap with the eventuality of its complement, which in turn is in conflict with the non-overlap imposed by the evidential, but this is just another way of saying that there is a semantic clash.

24. We can’t test it with *find*, since it only takes subjective predicates, which in English are all individual-level (Anand and Korotkova 2022; Pearson 2013) and as such can’t combine with temporal modifiers.

25. Koev (2017) incorrectly predicts that the evidential inference must project, yielding something along the lines of *I find the cake delicious—by the way, this is also what I heard* (which might be odd, but shouldn’t be infelicitous). Unless we can find another explanation of why the evidential is bad under a doxastic predicate, I maintain that the evidential component is locally accommodated, which in turn creates a conflict with the semantics of the embedding verb.

The bottom line of this excursion into the domain of attitude predicates is as follows. Certain predicates, such as English *notice* or *find*-verbs in many European languages, impose directness-like restrictions on how the subject acquired the complement proposition. They have been shown to ban indirect expressions in their scope, a property which [Korotkova and Anand \(2021\)](#) explain in semantic terms. While their account is world-based (doxastic predicates are treated as quantifiers over information states that semantically interact with evidentials in their scope), one may try to spell out the same intuition in terms of commitments or even events. Crucially, pragmatic approaches to evidentiality predict that no such ban would exist in the first place and that the evidential implicature would go away. Summing up, such data offer a novel tool for testing the predictions made by semantic vs. pragmatic theories of evidential meaning.

7 Outlook

Current debates on the proper semantics of evidentials in many ways resemble, on the one hand, the debates on epistemic modals as propositional operators vs. mere comments on propositional content (see overview in [Willer 2015](#)), and, on the other hand, the debates on the future as a temporal vs. modal operator (see overview in [Cariani 2021](#)).

The existing families of approaches have often been motivated by superficially different data and therefore focus on distinct features of evidential expressions: event-based accounts aim at establishing the (spatio)temporal underpinnings of evidential relations, world-based accounts examine the quantificational domains of evidential operators, and commitment-based accounts emphasize the illocutionary effects of evidential utterances. The overarching goals of this paper are to elucidate conceptual issues surrounding evidential meaning, to identify consequential differences between the existing accounts and to ascertain to what extent they can be brought to a common denominator. Purely event-based accounts are extensional and derive the evidential signal pragmatically. World-based accounts are intensional and hard-wire the evidential signal. Commitment-based accounts also hard-wire evidential restrictions, and while not always intensional, they can be made so. These two lesser-discussed dimensions, intensional vs. extensional and semantic vs. pragmatic, offer a new taxonomy of theories of evidential meaning. Purely event-based accounts portray a very different picture of what evidentials are, but world-based vs. commitment-based accounts are in fact not so different from each other. They both have sufficient machinery to handle most data and at present the choice between them seems to largely depend on one's assumptions about the semantics-pragmatics division of labor.

Drawing on various strands of research on modals, attitudes and discourse markers, the paper carefully goes over different empirical diagnostics that separately target various properties attributed to evidentials—intensionality, discourse-sensitivity, (in)directness—and discusses testable predictions made by different accounts of evidentials with respect to those properties. My hope is that a consistent application of these diagnostics will help adjudicate between different theories of evidential meaning, thus carving a way towards a better understanding of evidentiality as a semantic category.

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